

Abstract

This software component presents the implementation of an unsupervised anomaly detection system for a greenhouse Digital Twin, developed as part of the IJC308 Data Science Portfolio. The aim is to automatically identify unusual patterns in environmental sensor data that may indicate equipment malfunctions, sensor faults, or erroneous readings without requiring manually labelled training data.

The implementation employs an ensemble approach combining three complementary algorithms: Isolation Forest (Liu et al., 2008), One-Class SVM (Schölkopf et al., 2001), and Local Outlier Factor (Breunig et al., 2000). These methods represent distinct detection paradigms: tree-based isolation, boundary-based classification, and density-based neighbourhood analysis. This enables the system to capture different anomaly structures. Following Aggarwal and Sathe's (2017) recommendations for outlier ensembles, a majority voting mechanism flags anomalies only when at least two of three models agree, reducing false positives while maintaining sensitivity.

The system was evaluated on 10,081 sensor readings from an experimental greenhouse in Bleiswijk, Netherlands, identifying 227 anomalies (2.3%). Analysis revealed that detected anomalies correspond to operationally significant conditions including temperature extremes, high solar radiation, and strong wind events. Time-series visualisation confirmed that anomalies cluster around specific dates, suggesting genuine weather events rather than random noise.

Key limitations include the absence of ground truth labels for quantitative evaluation, a static contamination rate assumption, and no temporal modelling of sequential dependencies. Future work could incorporate sequence-based models such as LSTM autoencoders for time-aware detection, real-time deployment via AWS Lambda, and explainability features to identify which sensor values triggered each anomaly.